TIRTHA RUDRA SHETTIGAR

Electronics and Communication Engineering Student

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering student with a foundation in digital electronics, Verilog HDL, embedded systems, and electronic measurements. Hands-on project experience in RTL-based digital design, sensor interfacing, IoT systems, and machine learning applications. Interested in semiconductor product engineering, hardware validation, electronic testing, and data-driven engineering. Eager to apply analytical problem-solving and practical debugging skills in a semiconductor engineering internship.

EDUCATION

B.E. in Electronics and Communication Engineering 2023 – Present
Alva's Institute of Engineering & Technology, Moodabidri, Karnataka CGPA: 7.93

Pre-University Course – Science 2022 – 2023
Basaveshwar Science College, Bagalkot 75%

SSLC 2020 – 2021
Basaveshwara New High School, Bagalkot 78%

TECHNICAL SKILLS

Programming and HDL: C, C++, Python, Verilog HDL.
Digital Electronics: Digital logic, combinational and sequential circuits, finite state machines, CMOS fundamentals, RTL simulation, FPGA design fundamentals.
Semiconductor and EDA Tools: Intel Quartus Prime, ModelSim, Xilinx ISE, Cadence Virtuoso, Cadence Innovus.
Embedded Systems: Arduino, ESP32, Raspberry Pi, STM32, GPIO, ADC, PWM, UART, SPI, I2C, sensor interfacing.
Electronics and Measurements: Analog and digital electronics fundamentals, voltage and current measurement concepts, signal analysis, and electronic troubleshooting. Familiarity with laboratory instruments should be stated according to actual practical experience.
AI/ML and IoT: Python-based data analysis, regression, classification, clustering, sensor data acquisition, MQTT fundamentals, and IoT monitoring.

INTERNSHIP EXPERIENCE

Machine Learning Intern March 2026 – April 2026
Prodigy InfoTech *Remote*

- Worked on machine learning projects involving regression, classification, clustering, gesture recognition, and estimation.
- Developed project-based experience in house price prediction, K-Means customer segmentation, and SVM image classification.
- Explored hand gesture recognition and calorie estimation using Python-based machine learning workflows.

IoT & Embedded Systems Intern 2026
Upskill Campus / UniConverge Technologies *Remote*

- Completed six weeks of practical training in IoT and embedded systems.
- Developed obstacle avoidance and line-following robot projects involving sensor interfacing and embedded control.

SELECTED ENGINEERING PROJECTS

Verilog-Based Traffic Light Controller

- Designed a clock-driven traffic light controller using Verilog HDL and finite state machine concepts.
- Developed functional simulation workflows and explored synthesis and implementation using Intel Quartus Prime and ModelSim.

AI and IoT-Based Water Quality Monitoring System

- Developed a water quality monitoring workflow using pH, turbidity, TDS, and temperature sensors with Raspberry Pi and ADS1115.
- Explored Python-based data processing, machine learning-based risk analysis, IoT monitoring, and geographic risk visualization.

Smart Attendance System – TECHNOVATE 2K25

- Contributed to a smart attendance project integrating RFID-based identification and camera-based head-count verification.
- Secured first place at TECHNOVATE 2K25.

Embedded Robotics Projects

- Developed obstacle avoidance, line-following, and voice-controlled vehicle projects exploring sensor interfacing,

embedded control, and motor actuation.

RESEARCH & ACHIEVEMENTS

- **Research Paper:** "An Extensive Survey on IoT and AI/ML-Based Water Quality Monitoring," IEEE ICIET 2026, Paper ID 1062. Verify publication status and add the official paper link.
- **TECHNOVATE 2K25:** First Place for the Smart Attendance System.
- **Cognizant Technoverse Hackathon 2026:** Participated; include the specific achievement only if verified.
- **e-Yantra Robotics Competition, IIT Bombay:** Maze Solver Robot, Round 1.

ADDITIONAL TECHNICAL TRAINING

- MicroDegree – Python Full Stack Training.
- Pantech.AI – Embedded C and ARM Cortex-M4 Training.
- NPTEL – Verilog and FPGA / Signal Processing learning.